The Coffee Law: Information and Reasoning Ceilings for Tool-Using Agent Collectives

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***Abstract*—Multi-agent systems built from large language models (LLMs) are often expected to produce knowledge beyond any single agent through debate, critique and voting. We give information-theoretic ceilings on what such collectives can achieve, independent of how the agents work. First, the *Coffee Law*: for any number of agents, any communication protocol and any adaptive tool-query policy, the information the collective holds about an external truth Θ is at most $\min\{H(\Theta),\, I(\Theta; D_0) + \sum_t C_t\}$, where D_0 is the starting data and C_t is the Shannon capacity of round t's tool calls. Fano's inequality turns this into a floor on error and a lower bound on the number of tool rounds. Second, a *reasoning ceiling*: the information that computationally bounded agents can *use* is at most what they *hold*, with the gap equal to an expected KL divergence; computation, including debate, can raise usable information but never available information. Third, *computational bounds* on closing that gap: only a collective's total compute matters, parity requires reading every input, and under standard cryptographic assumptions some fully available information is unusable by any polynomial-time collective. Together the two ceilings account for published findings that extra test-time compute does not help on knowledge-intensive tasks, that debate helps on reasoning tasks, and that debate performs about as well as voting at equal compute. Every finite example is verified exactly in public code.**

***Index Terms*—multi-agent systems, large language models, tool use, mutual information, channel capacity, Fano's inequality, usable information**

I. Introduction

LLM-based agent systems combine many model instances that post to a shared board, debate, vote and call tools such as search engines, code interpreters and APIs [1]–[3]. A common expectation is that such a collective can know more than any member. This paper asks how much more, and from where.

We answer with two ceilings that hold for *any* agents—LLMs, people or programs—because the proofs never inspect the agents:

- **The Coffee Law** (Theorem 1). A collective can never know more about the world than its starting data plus what its tools can carry. Communication and internal computation add nothing to this budget; only starting data and informative tool calls raise it. The number of agents does not appear.
- **The reasoning ceiling** (Theorem 2). What computationally bounded agents can *use* is at most what they *hold*. Reasoning, including debate, closes this gap but never passes it.

Together they give the chain

$$\underbrace{I_{\mathcal{V}}(D_T \to \Theta)}_{\text{usable}} \le \underbrace{I(\Theta; D_T)}_{\text{available}} \le \underbrace{\min\{H(\Theta),\, B_T\}}_{\text{budget}}, \qquad (1)$$

with $B_T = I(\Theta; D_0) + \sum_{t<T} C_t$, where reasoning raises the left term and tools raise the right one. Our contributions are: (i) the capacity budget for tool-using collectives with adaptive queries, with a Fano error floor and a tightness example; (ii) the combination of available and usable information into one chain, with an exact expression for the reasoning gap; (iii) computational bounds on closing that gap, from above by total compute and from below unconditionally (parity) and conditionally (a hard-core bit); (iv) an account of several published empirical findings on debate and test-time compute (Section VII); and (v) exact numerical verification of every finite example.[1]

The ingredients—the chain rule, the data-processing inequality (DPI), channel capacity and Fano's inequality—are classical [4], [5]. The contribution is their assembly into a model of tool-using agent collectives and the resulting separation of information-bound from compute-bound tasks.

II. Model

All variables are discrete with finite support; logarithms are base 2. Θ is a fixed external truth. D_t is the collective's *full record* at the start of round t: the initial data D_0 (every agent's context and weights) plus every post, private state, query and tool result since. Each agent's state and the board are functions of D_t. In round t, fresh internal randomness ξ_t (sampling, scheduling) is drawn; the round's tool calls $Q_t = g_t(D_t, \xi_t)$ are issued under any policy g_t; results $O_{t+1} \sim K_t(\cdot \mid \Theta, Q_t, D_t)$ return; communication and computation use fresh randomness ζ_t; and

$$D_{t+1} = (D_t, \xi_t, Q_t, O_{t+1}, \zeta_t).$$

Assumptions. (A1) *No side channels*: Θ reaches the collective only through D_0 and tool results. (A2) *Truth-independent noise*: $\xi_t \perp (\Theta, D_t)$ and $\zeta_t \perp (\Theta, D_t, \xi_t, Q_t, O_{t+1})$. (A3) *Tools are channels from* Θ: given (Θ, Q_t, D_t), O_{t+1} is independent of ξ_t with law K_t; tools may be adaptive or stateful. (A4) *Fixed truth*: actions do not change Θ.

Capacity. For a query q and history d, $\theta \mapsto K_t(\cdot \mid \theta, q, d)$ is a channel with capacity $\mathrm{Cap}(K_{t,q,d}) = \max_{P_\Theta} I(\Theta; O)$. Let

[1] Code, proofs and checks: https://github.com/mbhatt1/coffeelaw.

$C_t := \sup_{q,d} \mathrm{Cap}(K_{t,q,d}) \le \log |\mathcal{O}_t|$, where $\mathcal{O}_t$ is the set of possible results in round t.

III. Conservation and Synergy

Proposition 1 (Conservation)**.** *If no tools are called ($C_t = 0$ for all t), then $I(\Theta; D_T) \le I(\Theta; D_0)$.*

This is the special case of Theorem 1 with no tools; it also follows directly from the DPI, since D_T is a function of D_0 and Θ-independent noise. Debate, critique and voting therefore add no information about Θ.

Proposition 2 (Synergy)**.** *Let B_1, B_2 be i.i.d. uniform bits and $\Theta = B_1 \oplus B_2$. Then $I(\Theta; B_1) = I(\Theta; B_2) = 0$ but $I(\Theta; B_1, B_2) = 1$.*

Pooling can make jointly held information accessible that no member can use alone; this is the failure of submodularity of $S \mapsto I(\Theta; S)$ studied in partial information decomposition [6]. The bit was present in D_0 throughout: emergence rearranges the budget without enlarging it.

IV. The Coffee Law

Theorem 1 (Coffee Law)**.** *Under (A1)–(A4), let $B_T := I(\Theta; D_0) + \sum_{t<T} C_t$. Then*

$$I(\Theta; D_T) \le \min\{H(\Theta),\, B_T\},$$

and the same bound holds for every output Y of the collective (answer, board, vote, any agent's state). For any estimator $\hat{\Theta}$ computed from the record, with $|\Theta| \ge 3$,

$$\Pr(\hat{\Theta} \ne \Theta) \;\ge\; \frac{H(\Theta) - \min\{H(\Theta), B_T\} - 1}{\log(|\Theta| - 1)}.$$

Proof. By the chain rule, $I(\Theta; D_{t+1}) - I(\Theta; D_t)$ equals

$$I(\Theta; \xi_t | D_t) + I(\Theta; Q_t | D_t, \xi_t) + I(\Theta; O_{t+1} | S) + I(\Theta; \zeta_t | S, O_{t+1}),$$

with $S = (D_t, \xi_t, Q_t)$. The first and last terms vanish by (A2); the second vanishes because Q_t is a function of (D_t, ξ_t). For the third, fix $S = s = (d, x, q)$. By (A3), given $S = s$ the pair (Θ, O_{t+1}) has $\Theta \sim p(\cdot|s)$ and $O_{t+1} \sim K_t(\cdot|\Theta, q, d)$, so $I(\Theta; O_{t+1} | S = s)$ is the mutual information of channel $K_{t,q,d}$ under input law $p(\cdot|s)$, at most its capacity and hence at most C_t. Averaging over s and summing over rounds gives $I(\Theta; D_T) \le B_T$; also $I(\Theta; D_T) \le H(\Theta)$. Outputs are functions of D_T and independent noise, so the DPI extends the bound to them. For the error floor, Fano's inequality [4] gives $H(\Theta|\hat{\Theta}) \le 1 + \Pr(\hat{\Theta} \ne \Theta) \log(|\Theta| - 1)$, and the DPI gives $H(\Theta|\hat{\Theta}) \ge H(\Theta) - \min\{H(\Theta), B_T\}$. □

Adaptivity is already accounted for: the query policy only selects the input law $p(\cdot|s)$ and the channel $K_{t,q,d}$, and capacity is the maximum over both. This mirrors the classical fact that feedback does not increase the capacity of a memoryless channel [7], [8] and the use of Fano's inequality for active-learning lower bounds [9].

TABLE I
Exact information about Θ after two tool rounds (budget $2C$).

Policy	$I(\Theta; D_2)$ (bits)	Budget $\min\{H, 2C\}$
Query bit 0, then bit 1	1.0620	1.0620
Adaptive, may repeat a bit	0.9820	1.0620
Either policy, plus a debate step	unchanged	—
Both bits, noise-free tool	2.0000	2.0000

Corollary 1 (Rounds needed)**.** *If Θ is uniform on 2^k values ($k \ge 2$), $I(\Theta; D_0) = 0$ and $C_t \le C$ for all t, then reaching $\Pr(\hat{\Theta} \ne \Theta) \le \varepsilon$ requires $T \ge ((1 - \varepsilon)k - 1)/C$ tool rounds, regardless of the number of agents or how they communicate.*

Corollary 2 (Agents do not appear)**.** *Synergy, debate, voting and the number of agents N are absent from B_T. More agents raise the budget only by issuing more or better tool calls, i.e. by raising C_t.*

Tightness. Let Θ be uniform on $\{0, 1, 2, 3\}$ and let a call $q \in \{0, 1\}$ return bit q of Θ through a binary symmetric channel with flip probability 0.1, so $C = 1 - h(0.1) \approx 0.531$ bits. Table I gives exact values. Querying both bits attains the budget; an adaptive policy that sometimes re-queries the same bit falls below it; a debate step adds nothing.

V. The Reasoning Ceiling

Holding information is not the same as being able to use it: the prime factors of N are fully determined by N, yet finding them may take enormous computation. Following Xu et al. [10], we model bounded agents by a predictive family.

Definition 1. A *predictive family* $\mathcal{V}$ is a set of predictors f mapping a record d to a distribution $f[d]$ on Θ, containing the constant prior predictor $f[d] = p_\Theta$. Define $H_\mathcal{V}(\Theta|D) := \inf_{f \in \mathcal{V}} \mathbb{E}[-\log f[D](\Theta)]$ and the *usable information* $I_\mathcal{V}(D \to \Theta) := H(\Theta) - H_\mathcal{V}(\Theta|D)$.

Xu et al. use the family's own no-input entropy as the baseline; with the prior predictor in $\mathcal{V}$, as assumed here, the baseline is $H(\Theta)$.

Theorem 2 (Reasoning ceiling)**.** (a) $0 \le I_\mathcal{V}(D \to \Theta) \le I(\Theta; D)$*, and the* reasoning gap *is*

$$I(\Theta; D) - I_\mathcal{V}(D \to \Theta) = \inf_{f \in \mathcal{V}} \mathbb{E}_D\big[D_{\mathrm{KL}}\big(p(\cdot|D) \,\|\, f[D]\big)\big].$$

(b) *Under (A1)–(A4), the chain* (1) *holds for the record D_T.*

(c) *There exist $\mathcal{V}$ and a deterministic g with $I(\Theta; g(D)) = I(\Theta; D)$ but $I_\mathcal{V}(g(D) \to \Theta) > I_\mathcal{V}(D \to \Theta)$. For every g, $I_\mathcal{V}(g(D) \to \Theta) \le I(\Theta; D)$.*

Proof. (a) The prior predictor gives $H_\mathcal{V}(\Theta|D) \le H(\Theta)$. For any f, the cross-entropy identity gives $\mathbb{E}[-\log f[D](\Theta)] = H(\Theta|D) + \mathbb{E}_D D_{\mathrm{KL}}(p(\cdot|D) \| f[D])$; take the infimum and subtract from $H(\Theta)$. (b) Part (a) and Theorem 1. (c) The last claim

is (a) followed by the DPI. For the strict example take Proposition 2 with $D = (B_1, B_2)$ and the "one-look" family $\mathcal{V}_1$ of predictors that read a single coordinate of the record. Each coordinate of D is independent of Θ, so $I_{\mathcal{V}_1}(D \to \Theta) = 0$. After one reasoning step $g(D) = (B_1 \oplus B_2, B_2)$ the first coordinate equals Θ, so $I_{\mathcal{V}_1}(g(D) \to \Theta) = 1 = I(\Theta; g(D)) = I(\Theta; D)$. □

Part (c) is the observation of [10] that usable information violates the DPI, placed in the collective setting.

Corollary 3 (Debate adds reasoning, not facts)**.** *In a closed collective every post is a function of D_0 and noise, so it adds no available information (Proposition 1). It can add usable information: an agent that posts an intermediate result hands others a computation outside their family. The gain is capped by $I(\Theta; D_0)$.*

Two regimes. A task is *information-bound* when the right inequality of (1) binds (forecasting, facts about the world, diagnosing a live system): only tools and better starting data help. It is *compute-bound* when the left gap is large (mathematics, proofs, code, planning from known facts): more reasoning helps. A code interpreter applied to data already held enlarges $\mathcal{V}$; a tool that observes the world raises C_t. In the noisy example of Table I, the record holds 1.0620 bits but a one-look agent can use only 0.2655 of them; the remaining 0.7965 bits are a reasoning gap that more observations cannot close.

Imperfect agents. Real agents fall further below the chain; the following are proved in the accompanying notes (footnote 1). For a decision with utility range U, acting on a belief $\hat{b}$ instead of the correct posterior b^* costs at most $2U\,\mathrm{TV}(b^*, \hat{b}) + \delta \le 2U\sqrt{(\ln 2/2)\, D_{\mathrm{KL}}(b^* \| \hat{b})} + \delta$, where δ is the agent's suboptimality given its own belief. A summary M of data D loses exactly $\mathbb{E}\, D_{\mathrm{KL}}(p(\cdot|D) \| p(\cdot|M)) = I(\Theta; D) - I(\Theta; M)$. And a message with zero information about Θ can still flip a decision: an agent with belief $(0.6, 0.4)$ in favor of the truth that gives weight α to a constant false claim answers wrongly once $\alpha > 1/6$.

VI. Computational Bounds on Reasoning

How much computation does closing the reasoning gap take? Fix a machine model (a word RAM) and let $\mathcal{V}_S$ be the predictors whose probabilities can be computed, and which can be sampled from, in at most S steps, reading one symbol of the record per step. Each $\mathcal{V}_S$ contains the prior predictor and $S \mapsto I_{\mathcal{V}_S}$ is nondecreasing and at most $I(\Theta; D)$. A collective's *total compute* is the sum over agents and rounds of their steps, with each board read or write counted as a step.

Theorem 3 (Compute ceiling)**.** *A predictor output by a collective with total compute S lies in $\mathcal{V}_{\kappa S}$ for a constant κ depending only on the machine model, so the collective's usable information is at most $I_{\mathcal{V}_{\kappa S}}(D \to \Theta) \le I(\Theta; D)$.*

Proof. One machine can run the agents step by step in a fixed interleaving, keeping each agent's memory and the board in separate regions, at $O(1)$ cost per simulated step; it outputs the same predictor in κS steps. □

The number of agents enters only through S: parallelism buys wall-clock time, not capability.

Lemma 1. *Sampling $\hat{\Theta} \sim f[D]$ succeeds with probability at least $2^{-\mathbb{E}[-\log f[D](\Theta)]}$. If $H(\Theta) \le 1$ and f achieves usable information δ, it succeeds with probability at least $2^{\delta - 1} \ge \frac{1}{2} + \frac{\ln 2}{2}\delta$.*

Proof. The success probability is $\mathbb{E}[f[D](\Theta)]$; apply Jensen's inequality to $-\log$, and use $2^{\delta-1} = \frac{1}{2}e^{\delta \ln 2}$. □

Theorem 4 (Compute lower bounds)**.** *(a) Let $D = (B_1, \ldots, B_n)$ be uniform bits and $\Theta = \bigoplus_i B_i$. Any predictor, or any collective counting all agents' reads, that reads at most $n - 1$ bits of D, even adaptively, has zero usable information; reading all n gives one bit. (b) Assume one-way permutations exist. For a one-way permutation f, uniform $x, r \in \{0,1\}^n$, $D = (f(x), r)$ and $\Theta = \langle x, r \rangle \bmod 2$, we have $I(\Theta; D) = H(\Theta)$, yet every polynomial-time family has $I_{\mathcal{V}_{\mathrm{poly}}}(D \to \Theta) \le \log(1 + 2\varepsilon(n))$ with ε negligible.*

Proof. (a) Conditioned on any transcript of fewer than n reads, the unread bits are uniform and independent, so Θ is independent of the transcript; the output is a function of the transcript and independent randomness, so Theorem 2(a) gives zero. Posts are functions of their authors' reads, so the bound covers the union of a collective's reads. (b) Θ is a function of D because f is a bijection. By the Goldreich–Levin theorem [11], every polynomial-time algorithm guesses Θ with probability at most $\frac{1}{2} + \varepsilon(n)$; by Lemma 1, every polynomial-time predictor has cross-entropy at least $1 - \log(1 + 2\varepsilon)$, and $H(\Theta) \le 1$. □

By Theorem 3, part (b) holds for any polynomial-time collective, however many agents it has. Part (a) shows splitting work can only reduce wall-clock time: N agents can each read n/N bits and post block parities, but the total is still at least n reads. Between these extremes no general efficient bound is possible without resolving open questions in complexity theory. The parity bounds and Lemma 1 are checked numerically in the code; part (b) is conditional.

VII. Consistency with Empirical Findings

The chain (1) accounts for several published results.

Information-bound tasks resist reasoning. Across 14 reasoning models on closed-book knowledge benchmarks, more test-time compute does not reliably raise accuracy and often increases hallucination [12]; the authors argue that compute-only scaling of a fixed model cannot add information about the answer, which is Proposition 1. Retrieval and tools, which raise C_t, are the standard remedy for knowledge-intensive tasks [2], [3], [13]. Intrinsic self-correction without external feedback likewise often fails to improve reasoning [14].

Compute-bound tasks benefit from debate. Multi-agent debate improves mathematical and strategic reasoning [1], consistent with debate closing a reasoning gap.

Debate is worth its reasoning gain. Debate does not reliably outperform self-consistency or ensembling and costs more [15], and most of its gains are attributable to majority voting [16]. If debate's value is computation, equal computation spent another way should do as well.

Two apparent tensions. Debate is also reported to improve *factuality* [1]. The facts were in the weights, i.e. in D_0, so debate made stored information usable without adding any. Debate is proved not to improve expected correctness for Bayesian agents [16]; such agents have an unrestricted $\mathcal{V}$, so their reasoning gap is zero and Theorem 2 predicts no gain.

Testable predictions. (P1) Accuracy versus the number of informative tool calls lies above the floor of Theorem 1. (P2) At matched compute, debate improves compute-bound tasks but not information-bound ones. (P3) Redundant or low-capacity tool calls add little, even when numerous.

VIII. Related Work

The closed-system case for multi-agent debate is proved via the DPI in [17], and debate is modeled as a belief martingale in [16]. Yang et al. [18] bound multi-agent performance by task uncertainty and a sum of per-agent channel informations, the same "$\min\{H, \sum \text{channels}\}$" shape with agents rather than tool calls as the channels. Usable information under computational constraints is due to [10]; synergy is studied in partial information decomposition [6]; directed information and feedback in [8]; and Fano-based lower bounds for active learning in [9]. To our knowledge, the per-call capacity budget for tool-using collectives with adaptive queries, and its combination with usable information, have not been stated before.

IX. Limitations

The bounds limit information about a fixed external truth, and do not cover outcomes the agents control (A4). The computational bounds are task-specific: Theorem 4 gives one unconditional and one conditional case, not a general formula for the compute a given gap requires. Assumption (A1) requires accounting for everything a model absorbed in training; a missed side channel invalidates the accounting. Results are for finite alphabets; a noiseless continuous-output tool has unbounded capacity. Tightness is shown by example, not in general. We report no new experiments; the predictions of Section VII are untested here.

X. Conclusion

A collective of agents can be clever, but it cannot know more than it has seen. The Coffee Law bounds what a tool-using collective can learn by its starting data plus the capacity of its tool calls; the reasoning ceiling bounds what it can use by what it holds; and reaching that ceiling costs computation that only total compute can buy, from linear in the input to beyond any efficient collective, depending on the task. Practically: if the answer is in the data, add reasoning; if it is not, add better observations of the world. Neither substitutes for the other.

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